

Aligned but Inert: Concept Bottlenecks in Dense Prediction, and How to Make Them Causal

Anonymous WACV Applications Track submission

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Abstract

Concept bottleneck models promise intrinsic interpretability: route the prediction through a human-readable concept layer, and the concept activation is the explanation. We show that transplanting this idea into dense prediction fails silently, diagnose why, and repair it. We build a concept-routing bottleneck that assigns every encoder token to one of K clinically-labeled experts, and train it for 3D brain-tumor segmentation. By every measure this literature reports, it succeeds: routing tracks the clinical concepts at CAS_{fg} up to 0.96 and separates necrosis, edema and enhancing tumor at AUROC 0.88, against 0.74 for the strongest post-hoc attribution. Yet the routing is causally inert. Forcing tokens through a different expert changes 0.2% of voxel labels, and zeroing the entire bottleneck costs 1.5 Dice. Alignment metrics cannot detect this, because alignment is compatible with a bottleneck the model never consults—and the alignment loss trains precisely what those metrics score. We trace the failure to two leaks that encoder–decoder architectures invite, a residual path through the module and skip connections around it, and rule out the obvious explanation: the experts are not degenerate (0.9 pairwise divergence). A linear probe on an encoder trained with no concept supervision already reaches 0.81, so concept-discriminability gains over saliency baselines are weaker evidence than they appear. Attempts to force dependence by removing the decoder’s alternatives fail—the optimizer compensates. Making the routing an additive term of the output instead succeeds: causal control rises from 0.011 to **0.636** [0.621, 0.651] label-flip under intervention ($p < 10^{-28}$), removing the routing term costs up to 99% Dice, and the price is ≈ 1 Dice point. **A concept layer is an explanation only if the model consults it, and showing that requires intervening, not correlating.**

1. Introduction

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2. Related Work

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3. Method

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4. Diagnostic Protocol

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5. Experiments

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6. Discussion and Limitations

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